

SpaceX Falcon 9 First-Stage Landing Prediction

Data collection • EDA • SQL • Folium • Plotly Dash • Machine Learning

GitHub repository

Completed capstone notebooks and Python files are available in my GitHub repository:

<https://github.com/trandaitrang1210-sudo/Applied-Data-Science-Capstone>

Executive Summary

- Goal: predict whether the Falcon 9 first stage lands successfully, a key driver of launch reusability and cost.
- Data collection combines the SpaceX REST API and web scraping; the course API workflow yields 90 Falcon 9 records and 17 features after filtering.
- EDA uses Pandas/NumPy, SQL, Matplotlib/Seaborn, Folium, and Plotly Dash to study launch sites, payload, orbit, flight history and landing outcomes.
- Predictive models include Logistic Regression, SVM, KNN and Decision Tree with GridSearchCV hyperparameter tuning.
- Reference course-compatible results show Decision Tree with the strongest validation score (~ 0.889); test-set accuracy is 83.33% in the graded workflow.

Introduction & Business Question

- SpaceX reduces launch cost by recovering and reusing the Falcon 9 first stage.
- The analytical question is: given launch features such as payload mass, orbit, launch site, booster configuration and flight history, can we predict first-stage landing success?
- A reliable prediction can support competitive launch-cost analysis and bidding decisions.
- Target variable: Class = 1 for successful landing and Class = 0 for unsuccessful landing.

Core analytical path

Collect → Clean/Encode → Explore → Visualize → Build models → Tune → Evaluate → Conclude

Methodology: Data Collection & Wrangling

- API: `requests.get()` retrieves SpaceX launch JSON; `pandas.json_normalize()` converts nested JSON into a dataframe.
- Filtering retains Falcon 9 launches. The course-compatible API dataset contains 90 records and 17 selected features.
- Web scraping: BeautifulSoup parses the Falcon 9 / Falcon Heavy launch table and converts HTML rows into a dataframe.
- Wrangling: inspect missing values, fill/handle missing payload values, derive landing outcome Class, and retain modeling features.
- Categorical fields such as Orbit, LaunchSite, LandingPad and Serial are one-hot encoded before modeling.

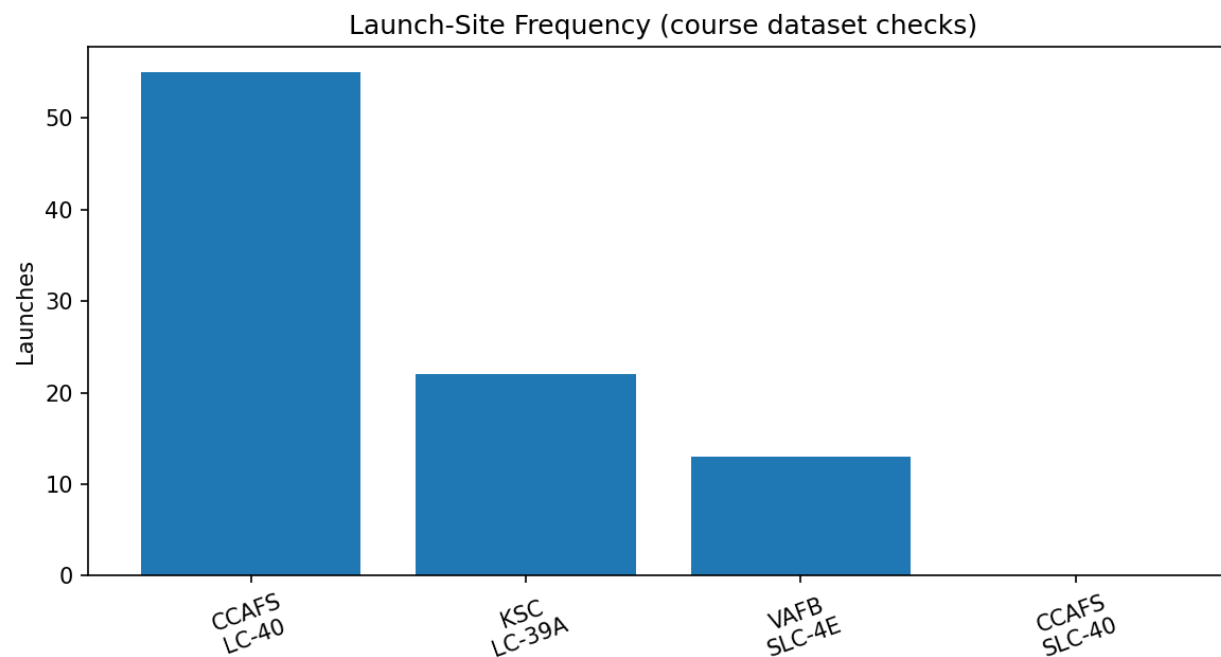
Methodology: EDA & Interactive Visual Analytics

- EDA with Pandas/NumPy: launch-site frequencies, orbit frequencies, mission outcomes and success rates.
- SQL: SELECT, DISTINCT, WHERE, GROUP BY, COUNT, SUM, AVG, MIN, BETWEEN and date filtering answer structured business questions.
- Matplotlib/Seaborn: scatter, categorical, bar and line plots examine FlightNumber, PayloadMass, LaunchSite, Orbit and Class.
- Folium: launch-site markers, success/failure marker clusters and distance-to-proximity analysis.
- Plotly Dash: launch-site dropdown, payload-range slider, success pie chart and payload-vs-outcome scatter plot.

Methodology: Predictive Analysis

- Prepare feature matrix with numerical variables and one-hot encoded categorical variables.
- Standardize features with StandardScaler.
- Split data into training and test sets; the graded workflow test set contains 18 records.
- Train Logistic Regression, Support Vector Machine, Decision Tree and K-Nearest Neighbors.
- Use GridSearchCV to select hyperparameters; compare validation scores, test accuracy and confusion matrices.

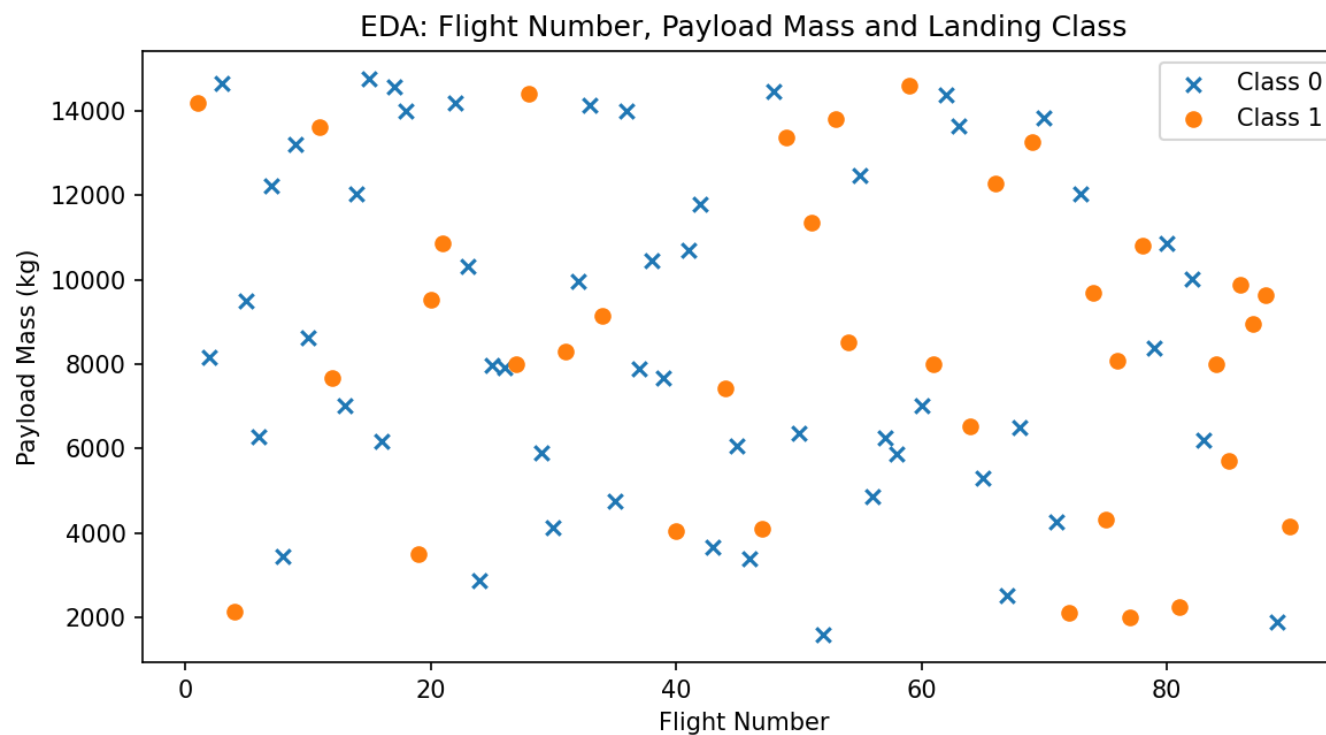
EDA Result: Launch Sites



Interpretation

- CCAFS LC-40 is the dominant launch site in the course dataset check (55 launches).
- Launch-site frequency is highly uneven, so site identity can carry predictive information.
- Site effects should be interpreted together with payload, orbit and flight chronology.

EDA Result: Flight History, Payload & Landing Class



Key pattern

- Later flight numbers tend to contain more successful landings.
- Payload mass alone does not perfectly separate success and failure.
- Interactions among flight history, orbit and launch site motivate multivariate classification.

EDA with SQL: Queries & Results

Launches from CCAFS LC-40

Result: 55

```
SELECT COUNT(*) FROM SPACEXTBL WHERE Launch_Site='CCAFS LC-40';
```

Minimum payload mass

Result: Minimum value returned by dataset

```
SELECT MIN(PAYLOAD_MASS__KG_) FROM SPACEXTBL;
```

Total payload mass

Result: Aggregate total

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL;
```

Mission outcomes by site

Result: Count per site

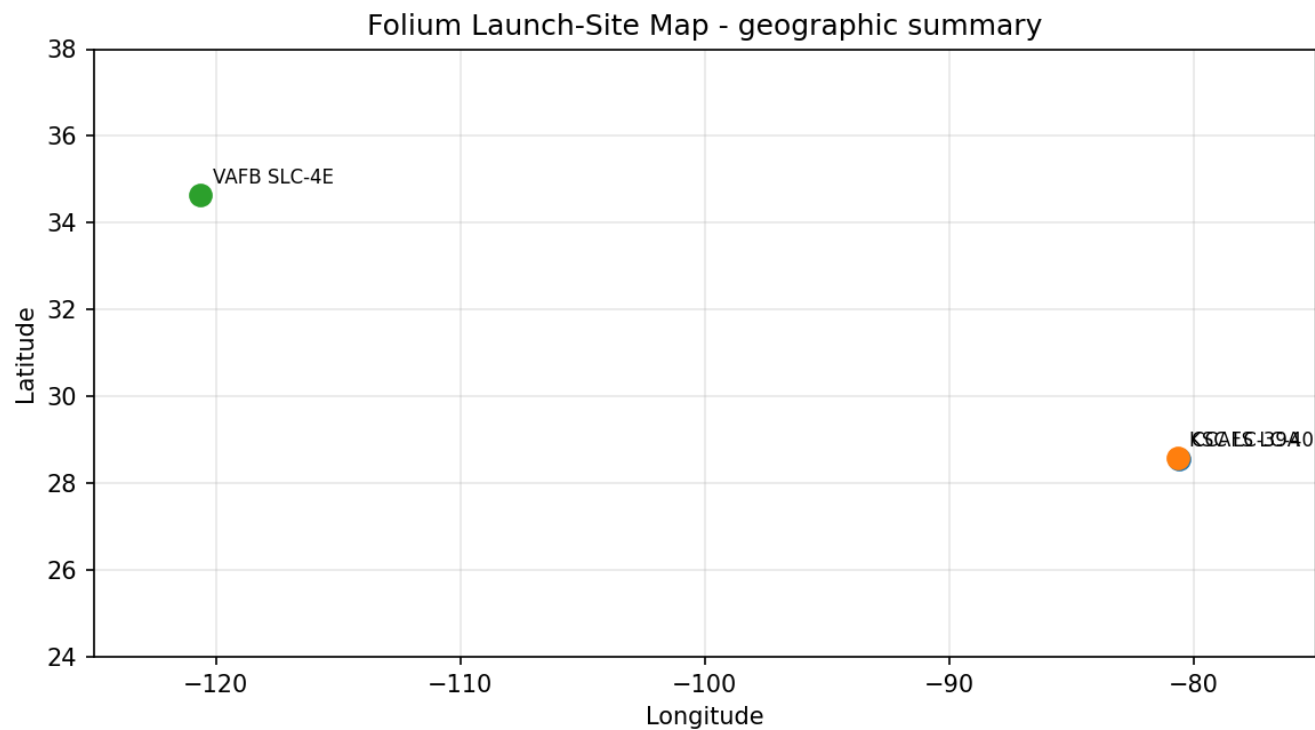
```
SELECT Launch_Site, COUNT(Mission_Outcome) FROM SPACEXTBL GROUP BY Launch_Site;
```

GEO orbit launches

Result: 1

```
SELECT COUNT(*) FROM SPACEXTBL WHERE Orbit='GEO';
```

Folium Map Results



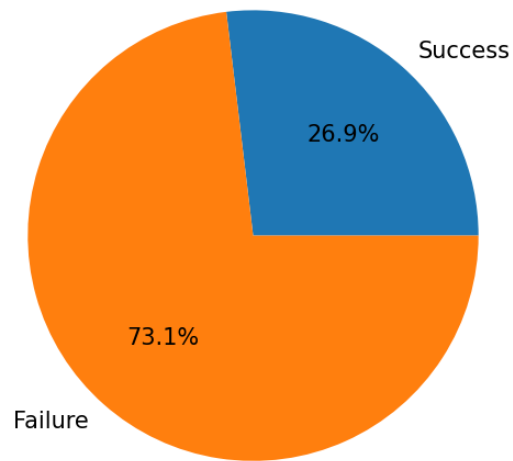
Map findings

- Launch sites are concentrated on U.S. coasts.
- Folium MarkerCluster separates individual launch records while reducing map clutter.
- Color-coded markers distinguish successful and failed landings.
- Distance lines to coastline, roads, railways and cities add geographic context for launch operations.

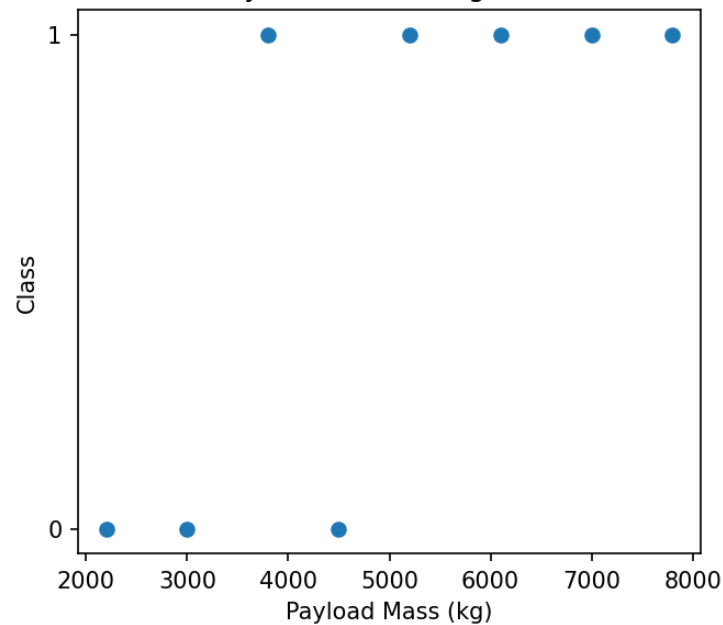
Plotly Dash Results

Plotly Dash - reconstructed dashboard view

CCAFS LC-40 Outcome Share



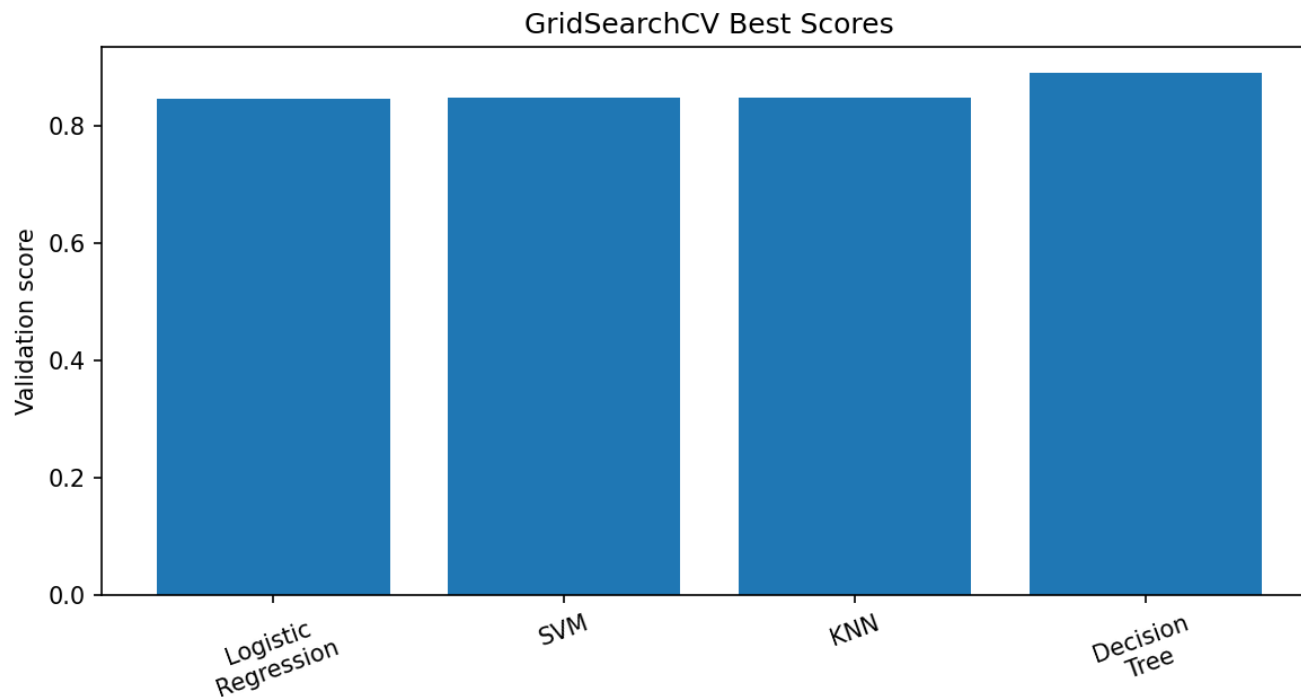
Payload vs. Landing Class



Interactive insights

- Dropdown filters all sites or one launch site.
- Payload slider restricts the payload-mass range.
- For CCAFS LC-40, the reference dashboard shows about 73.1% failed launches (Class 0).
- Scatter view helps compare payload mass with landing outcome and booster category.

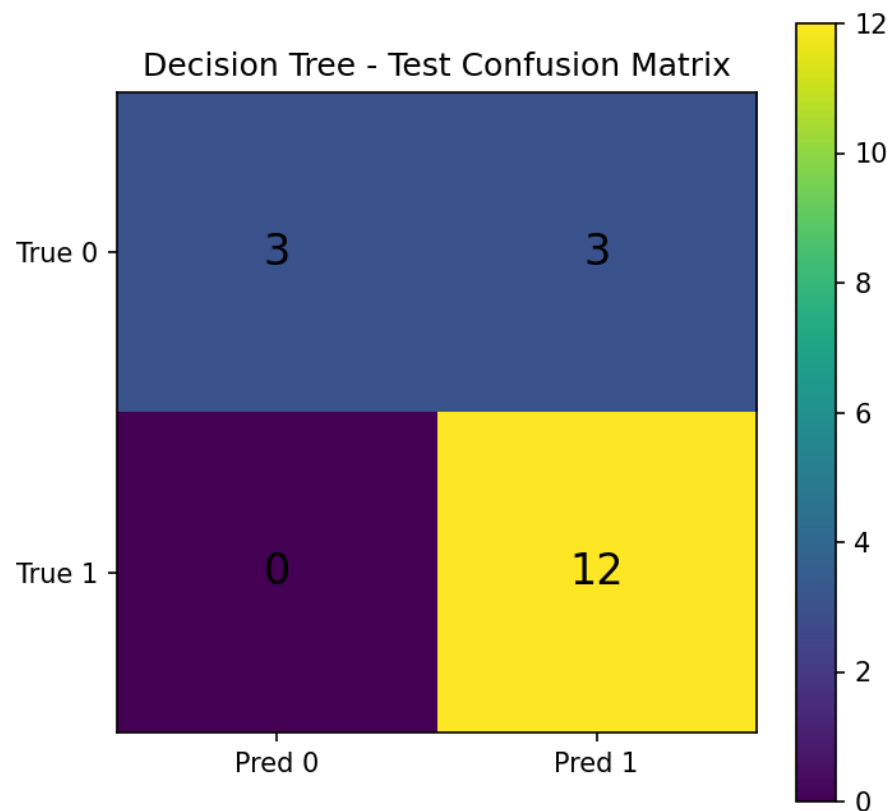
Predictive Analysis Results: Model Comparison



GridSearchCV scores

- Decision Tree: 0.8893
- KNN: 0.8482
- SVM: 0.8482
- Logistic Regression: 0.8464
- Decision Tree has the strongest validation score in the reference course workflow.

Predictive Analysis Results: Test Performance



Decision Tree test result

83.33% accuracy

- Test sample size: 18 records.
- 83.33% corresponds to 15 correct predictions out of 18.
- The small test set means a few records can noticeably change the reported accuracy.
- Use validation score and confusion matrix together rather than relying on accuracy alone.

Conclusion & Innovative Insights

- Landing success improves with launch experience: flight chronology contains useful signal.
- Payload mass is informative but does not determine landing success by itself; its effect interacts with orbit and launch site.
- Geospatial analysis shows launch facilities are coastal and near transport infrastructure, consistent with operational and safety constraints.
- Interactive Dash analysis makes site-specific success rates and payload effects easier to inspect than static summaries.
- Among the four classifiers, Decision Tree provides the strongest cross-validation score in the reference workflow; the graded test workflow reports 83.33% accuracy.
- Recommendation: use multiple launch attributes together and continue updating the model as newer launch records become available.

Submission Checklist

Included in this revised PDF

- Executive Summary
- Introduction
- Data Collection & Wrangling Methodology
- EDA & Interactive Visualization Methodology
- Predictive Analysis Methodology
- EDA Visualization Results
- EDA with SQL Queries/Results
- Folium Map Results
- Plotly Dash Results
- Predictive Analysis Results
- Conclusion & Innovative Insights

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